

Individual Reflective Essay: Digitization

The Shared Foundational Elements Between Digitizing

Audiovisual (AV) materials and Bitmapped Images

Name: Lucas Rosendall | SI581

Date: March 27th, 2025

Introduction

The digital is often a proxy for something else. “Digital Culture,” “Digital Media,” “Digital Humanities,” “Digital Preservation” largely leave *the digital* itself unquestioned. These are all legitimate sources of inquiry that sustain the digital through attribution and sociomaterial instantiation - the digital as a means or gateway to something else. But what is the digital in and of itself? Rather, if we refuse to confine it to its negation, namely to refuse to see it only in relation to the analog, what might be its positive existence? The digital as subcontract, or with something else acquiring semi-control over its theorization, risks conceptual invisibilization. We should maintain the digital and the analog as equal modes of habitation. But where might the qualitative nature of material give way to the discretizing abstractions of symbols and numbers? We can think of both the analogue and the digital as forms of mediation, one requiring continuous variation (analog) and the other requiring discrete units (digital). Or moreover, analog like an analogous thing, a correspondence with another, a transfer or carrying over – the carved groove on a vinyl record as the physical analog of the sound wave – existing on a continuous spectrum of infinite gradation. Or the digital, like your fingers or toes, countable numerics or digits, a sequence or string of individual numbers – the discrete values of an image's bits (depth) and their correspondence (or approximation) to the continuous original – like lossless pieces (characters, pixels) of arranged reality. Working from this viewpoint of mutual conditioning, the shared mediation of the analog and the digital, this reflection builds

on scholarship that foregrounds the best practices for digital imaging and the digitization of Audiovisual (AV) material in order to investigate how a medium might represent reality.

A Dialectic: The Digital and The Analog

Edmondson (2004) rightly tells us that content is shaped by carrier and context; obligingly, he probes the long duration of media archaeology through the vertical logic of mediums that come to delineate onto-epistemological messaging (chief among them being the three to four minute pop song and the standard playing time of the edison cylinder or the 78 rpm disk). But lets take for instance the IASA-TC 06 (2019) conception of the conventional videotape. This hallmark of physical substrate, our notable and courted ribbon of magnetically sensitive material, can carry either analogue (carrier and processing interdependence) or digital (bitstreams formatted independently) signals. Throughout the twentieth century and into the twenty-first form is not so easily detached from its materiality. Digital videotape in its media-dependent form remains susceptible to a carrier's physical or proprietary logic and therefore requires significant resources for transfer (playback equipment, recording systems, systems to monitor metrics and support quality review, and skilled personnel). In fact, IASA-TC 06 (2019) classification between media-dependent and file-based, independent media offers a composited and stratified picture as we move from one class to another. Generally, file formats (Matroska, MXF, or your Marketplace wrappers) liberate media from their form only to interpose a new medium altogether, a medium that comes to inform the logics of the present, namely digital computation itself. It is critical to recall that only through layers of staging can anything inside a computer be considered an object. The analog image, for example, is inverted into digital symbols, but in turn will oftentimes invert to real analogicity. This dialectical interlacing between the digital and the analog is, in my eyes, an important thing for our field to reckon with in detail, especially as it concerns the metaphors that make up archival practice.

Conclusion

These units on digitization call up for me the ontological status of the analog. What emerges across weeks nine and ten of our semester is not only the technical requirements – the accuracy of the capture

through sampling rate or bit depth (signal resolution) – or the completeness of preservation and the importance of a transparent workflow – through the encoding of technical and contextual metadata (secondary information) and sound project planning and development – but also the onto-epistemological shift from the qualitative (analogicity) to the quantitative (digitality). Libraries and archives currently face this maddening split, this radical rupture resemantifying our world. Edmondson's (2004) mournful display memorializing the aesthetic and sensory pleasures of the analog is not innocent. Indeed, it is a blatant disavowal of the digital as other and inferior, an analog chauvinism still prevalent and pervasive in our changing profession. I am by no means a digital evangelist, and recognize such loss, but such uniform discourse works to facilitate a catastrophic ontological arrangement – one that obscures discreteness underneath inherent preferences for continuous variation. Meanwhile, the digital has actively remade our semantic world; “preservation” as “digital preservation,” passing through the grip of its mediation, and translated into formal or syntactical rules called “Standards” and “Guidelines.” But what is the digital, really? Its empty, formal, and logical operations are the dominant medium of cultural life. The archive is currently being translated into its terms. To step back from the edge, our profession must draw on philosophy, media studies, design research, and epistemology to really reckon with digitization. And we should do this before our categories begin to buckle under the weight of the world they once described. The digital is a real space that does come before or after some integral body. Rather, thinking must be thought in relation to the digital.

References

Acker, A. (2025). *Archiving machines: From punch cards to platforms*. MIT Press.

Conway, Paul. *SI 581 Standards for Audio & Moving Image (AMI) Digitization*. 2026.

Conway, Paul. *Making Sense of Digitization Standards*: 2026.

Edmondson, Ray. *Audiovisual Archiving: Philosophy and Principles*; 2004.

Fazi, M. B., Alexander R. Galloway, Handelman, M., & Weatherby, L. (2025). *Digital theory*

International Association of Sound and Audiovisual Archives. (2019). *Guidelines for the preservation of video recordings (IASA-TC 06, Part B: Video signal, preservation concepts, and target formats)*.

Jones, Jimi, and Marek Jancovic. “‘Lossless’: The Materiality of Archival Video Format Standards.” *The Future of Memory*, University of Illinois Press, 2025, pp. 37–70. A History of Lossless Format Standards in the Moving Image Archive. JSTOR, <https://www.jstor.org/stable/10.5406/jj.29556574.6>.